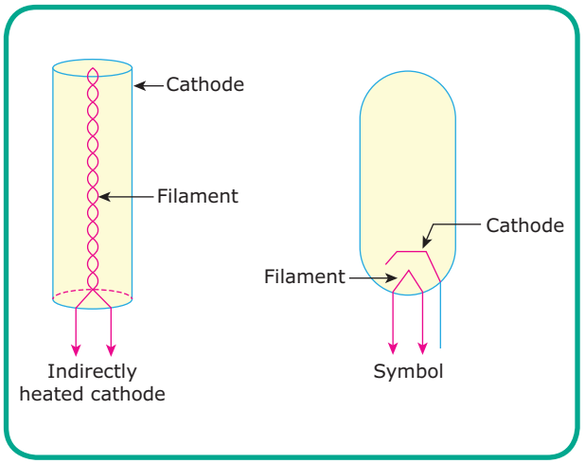


Figure 3.7 **Thermionic emission**

2. **Photo electric emission:** In this method, energy of light is falling upon the metal surface is transferred to the free electrons within the metal to enable them to leave the surface as shown in Figure 3.8. The greater the intensity (i.e., brightness) of light beam falling on the metal surface, the greater is the photoelectric emission.

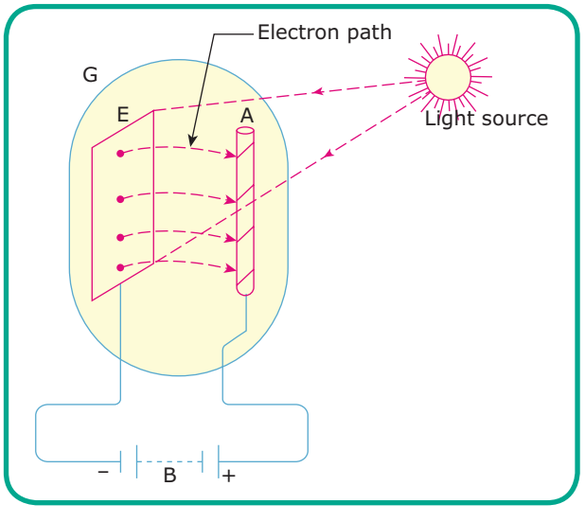


Figure 3.8 **Photo electric emission**

3. **Secondary Emission:** In this method, a high velocity beam of electrons strike the metal surface and causes the electron emission out of the surface as shown in Figure 3.9.

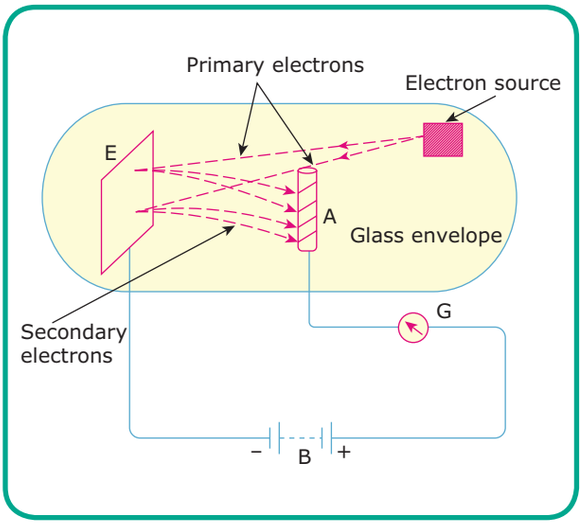


Figure 3.9 **Secondary emission**

4. **Field Emission:** In this method, a strong electric field (i.e., high positive voltage)is applied at the metal surface which pulls the electrons out of metal surface because of the attraction of positive field. The stronger the electric field, the greater is the electron emission as shown in Figure 3.10.

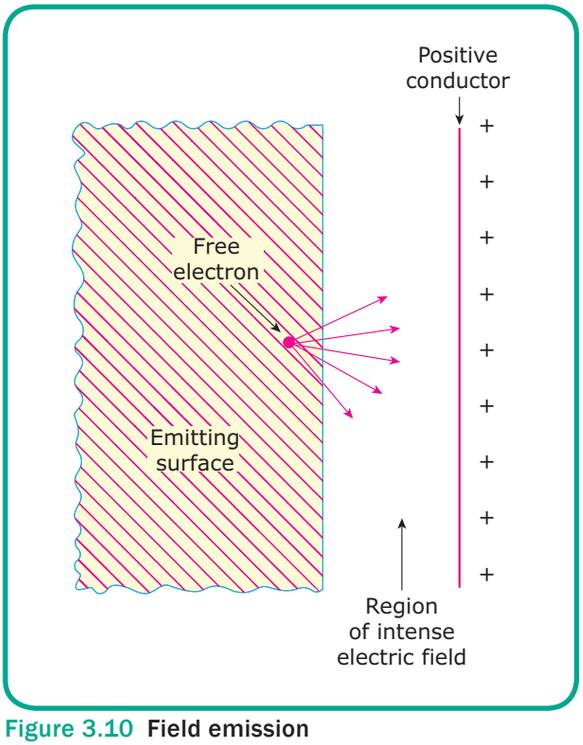


Figure 3.10 **Field emission**

ELEC_EQUIP_Chapter03.indd 66 11/10/2020 11:36:40 AM